

Commutative Algebra & Algebraic Geometry

Exercise 7

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7.1 Problem

(1) Let $k = \mathbb{C}$, determine which of the following is Zariski closed, and if it is compute its ideal:

(i) $X = \{(a, a^2) \mid a \in \mathbb{C}\},$

(ii) $X = \{(a, \sin a) \mid a \in \mathbb{C}\}.$

(2) Let A be a fg k -algebra, and let $X = \text{Spec}(A)$. Let $Y, Z \subseteq X$ be closed, then show

(i) $|Y \cup Z| = |Y| \cup |Z|,$

(ii) $|Y \cap Z| = |Y| \cap |Z|,$

(iii) Y is connected iff $|Y|$ is connected,

(iv) Y is irreducible iff $|Y|$ is irreducible.

(1)

(i) Clearly we have $X = Z(y - x^2)$ so X is closed. Its ideal is then

$$\mathcal{I}(X) = \sqrt{(y - x^2)} = (y - x^2),$$

since $y - x^2$ is irreducible as if $y - x^2 = f(x, y)g(x, y)$ then one of f, g must have degree 0 over y and the other degree 1. So $y - x^2 = p(x)g(x, y)$, but we must have that p is a constant since y has no x component in $y - x^2$.

(ii) Here X is not closed. If it were, then it would be contained in some proper $Z(f)$ for some $f \in \mathbb{C}[x, y]$. But if $f(a, \sin a) = 0$ for all $a \in \mathbb{C}$ then f must be zero, since $\sin x$ is not an algebraic function of x . Then $Z(f) = Z(0) = X$ a contradiction.

(2)

(i) We know that for $Y \subseteq X$ closed, $S \subseteq Y$ is relatively closed iff S is closed in X . Thus $p \in Y$ is a closed point in Y iff it is a closed point in X . Thus

$$|Y| = |X| \cap Y.$$

Since $Y \cup Z$ is closed, we see

$$|Y \cup Z| = |X| \cap (Y \cup Z) = (|X| \cap Y) \cup (|X| \cap Z) = |Y| \cup |Z|.$$

(ii) Similar to above, we have

$$|Y \cap Z| = |X| \cap (Y \cap Z) = (|X| \cap Y) \cap (|X| \cap Z) = |Y| \cap |Z|.$$

(iii) Since Y is closed, it is of the form $Z(I)$ for a radical ideal $I \leq A$. We know $Y = Z(I)$ is homeomorphic to $\text{Spec}(A/I)$, and A/I is a reduced fg k -algebra, so it is sufficient to show that X is connected (resp. irreducible) iff $|X|$ is.

Recall that $|X|$ is the maximal spectrum of A , with closed sets given by

$$Z_m(I) = Z(I) \cap |X| = \{\mathfrak{m} \leq A \text{ maximal} \mid I \leq \mathfrak{m}\}.$$

Then we have that

$$Z_m\left(\sum I_i\right) = Z\left(\sum I_i\right) \cap |X| = \bigcap_i Z(I_i) \cap |X| = \bigcap_i Z_m(I_i),$$

and

$$Z_m(IJ) = Z(IJ) \cap |X| = (Z(I) \cup Z(J)) \cap |X| = Z_m(I) \cup Z_m(J).$$

We make the following claims: $Z_m(I) = |X|$ iff $Z(I) = X$ and $Z_m(I) = \emptyset$ iff $Z(I) = \emptyset$. Both “if” directions are clear. The latter equivalence is also clear since $Z_m(I) = \emptyset$ means I is not contained in any maximal ideal and thus it cannot be a proper ideal, so $Z(I) = \emptyset$. For the former “only if” direction, let $\mathfrak{p} \in X$ then for every maximal extension $\mathfrak{p} \leq \mathfrak{m}$ we have $I \leq \mathfrak{m}$. Then

$$I \geq \bigcap_{\mathfrak{p} \leq \mathfrak{m}} \mathfrak{m} = \text{Jac}(\mathfrak{p}) = \mathfrak{p},$$

where the final equality is due to A being a Jacobson ring as a fg k -algebra, so $\mathfrak{p} \in Z(I)$ and $Z(I) = X$ as desired.

If X is disconnected, then X can be written as the disjoint union of two non-empty closed subsets. Say $X = Z(I) \cup Z(J)$, then $|X| = Z_m(I) \cup Z_m(J)$ are disjoint. And $Z_m(I), Z_m(J)$ are nonempty because we showed this means $Z(I)$ or $Z(J)$ is empty.

Conversely if $|X|$ is disconnected, so $|X| = Z_m(I) \cup Z_m(J)$ is a disjoint union of non-empty closed subsets. So $|X| = Z_m(IJ)$ and thus we showed $X = Z(IJ) = Z(I) \cup Z(J)$. This is a disjoint union since $Z_m(I) \cap Z_m(J) = Z_m(I + J) = \emptyset$ so $Z(I) \cap Z(J) = Z(I + J)$ is empty as well as we showed.

- (iv) If X is reducible, then $X = Z(I) \cup Z(J)$ for $Z(I), Z(J)$ proper. Then $|X| = Z_m(I) \cup Z_m(J)$ and these are proper as $Z(I), Z(J)$ are proper ($Z_m(I) = |X|$ iff $Z(I) = X$), so $|X|$ is reducible. Conversely if $|X| = Z_m(I) \cup Z_m(J) = Z_m(IJ)$ then $X = Z(IJ) = Z(I) \cup Z(J)$ and these are proper as $Z_m(I), Z_m(J)$ are proper, so X is reducible.

7.2 Problem

- (1) Let X be a topological space, and $\mathcal{F}: \text{Op}(X)^{\text{op}} \rightarrow \text{Alg}_k$ be a subpresheaf of $\text{Fun}(\bullet, k)$. Find an example of $\mathcal{F}$ satisfying the localization condition but not the sheaf condition. And similarly find a sheaf $\mathcal{F}$ not satisfying the localization condition.
- (2) Let X be a topological space and $\{B_i\}$ a basis for the topology. Show that to define a structure sheaf $\mathcal{O}$ on X it is enough to define it on the basic open sets so that $\mathcal{F}$ satisfies the axioms on the basic open sets (and also necessary).
- (3) Prove that SWF_k admits all colimits.
- (4) Show that in SWF_k , $(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2})$ is the product of $(\mathbb{A}^1, \mathcal{O}_{\mathbb{A}^1})$ with itself. Deduce that the forgetful functor $\text{SWF}_k \rightarrow \text{Top}$ is not a right-adjoint.
- (5) Find an SWF which is not a variety.
- (6) Fix two SWFs $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$. Find a continuous map $X \rightarrow Y$ which does not lift to a

morphism of SWFs. And find a morphism $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ which is not an isomorphism, but $X \rightarrow Y$ is a homeomorphism.

- (1) Let $X = \mathbb{R} = k$ and equip it with the discrete topology. Then for every $S \subseteq \mathbb{R}$ let $\mathcal{O}(S)$ be the set of functions $S \rightarrow \mathbb{R}$ which are continuous in the usual topology. This satisfies the localization condition: for $f \in \mathcal{O}(S)$ then $D_S(f)$ is open (as all sets are open) and $1/f$ is continuous over it in the usual topology (as the composition with the continuous $D_S(f) \rightarrow \mathbb{R}^\times \xrightarrow{\bullet^{-1}} \mathbb{R}^\times$) so $1/f \in \mathcal{O}(D_S(f))$. But it is not a sheaf. For one, clearly $\mathcal{O}(X)$ is not all of $\text{Fun}(X, \mathbb{R})$ since not all functions are continuous in the usual topology. But let $f: X \rightarrow \mathbb{R}$ not be in $\mathcal{O}(X)$, then $\{x\}_{x \in X}$ is an open cover of X and $f_x: \{x\} \rightarrow \mathbb{R}$ given by $f_x(x) = f(x)$ is continuous so $f_x \in \mathcal{O}(\{x\})$. If $\mathcal{O}$ were a sheaf, then there would be a $g \in \mathcal{O}(X)$ such that $g|_{\{x\}} = f|_{\{x\}}$ for all x , i.e. $g(x) = f(x)$ so $g = f$ identically. But we assumed $f \notin \mathcal{O}(X)$, so $\mathcal{O}$ is not a sheaf.

The presheaf $\text{Fun}(\bullet, k)$ is itself a sheaf, independent of the underlying topology. But it is not necessarily local. For example take $X = \mathbb{R}$ with the trivial topology, so $\text{id} \in \text{Fun}(X, k)$. But $D_X(\text{id}) = X - 0$ is not open, so $\text{Fun}(\bullet, k)$ is not local.

- (2) Let $U \subseteq X$ be open and let $\{B_i\}$ be a cover of U with basic open sets. Then define

$$\mathcal{O}(U) = \{f: U \rightarrow k \mid f|_{B_i} \in \mathcal{O}(B_i) \text{ for all } i\}.$$

We claim that this is independent on the open cover. Indeed, if $\{B_j\}$ is another open cover and $f|_{B_i} \in \mathcal{O}(B_i)$ for all i , we want to show $f|_{B_j} \in \mathcal{O}(B_j)$ for all j .

Let $\{B_k^{ij}\}$ be an open cover for $B_i \cap B_j$ for an i, j , so $\{B_k^{ij}\}_{i,k}$ covers B_j and $\{B_k^{ij}\}_{j,k}$ cover B_i . So there are $f_{ijk} \in \mathcal{O}(B_k^{ij})$ such that $f_{ijk}|_{B_k^{ij}} = f|_{B_k^{ij}}$. Since $\{B_k^{ij}\}_{i,k}$ cover B_j this means that there is a $g \in \mathcal{O}(B_j)$ such that $g|_{B_k^{ij}} = f|_{B_k^{ij}}$ for all i, k . And thus $g = f|_{B_j}$, so $f|_{B_j} \in \mathcal{O}(B_j)$ as desired.

In particular this means that for $B = B_i$ basic, with the open cover $\{B\}$ we get that our new definition of $\mathcal{O}(B)$ agrees with the given $\mathcal{O}(B)$.

Now say $U \subseteq V$ and $f \in \mathcal{O}(V)$. If we let $\{B_i\}$ cover U and extend it to a covering of V , we see that $f|_{B_i} \in \mathcal{O}(B_i)$ for all i so by definition $f|_U \in \mathcal{O}(U)$ as well, meaning $\mathcal{O}$ is a presheaf.

And if $\{U_i\}$ is an open cover of U and $f: U \rightarrow k$ has $f|_{U_i} \in \mathcal{O}(U_i)$ for all i , then let B_j^i cover U_i so that $\{B_j^i\}_{i,j}$ covers U then by definition $f|_{B_j^i} \in \mathcal{O}(B_j^i)$ for all i, j . So $f \in \mathcal{O}(U)$, and thus $\mathcal{O}$ is a sheaf.

If $f \in \mathcal{O}(U)$, cover U by B_i so that

$$D_U(f) = \bigcup_i D_{B_i}(f)$$

is open. Cover $D_{B_i}(f)$ by B_i^j so by assumption $1/f|_{B_i^j} \in \mathcal{O}(B_i^j)$, and thus by definition $1/f|_{D_{B_i}(f)} \in \mathcal{O}(D_{B_i}(f))$. These cover $D_U(f)$ so $1/f \in \mathcal{O}(D_U(f))$ showing that $\mathcal{O}$ is local, and thus a structure sheaf.

- (3) I unfortunately don't have time to prove these constructions. For a coproduct $Y = \coprod_i X_i$, it is sufficient to define the structure sheaf on the basic open sets of the coproduct, which are just the open subsets of X_i . For $U \subseteq X_i$ define

$$\mathcal{O}_Y(U) = \mathcal{O}_{X_i}(U).$$

Now if $\pi: X \rightarrow Y$ is a quotient map, there is a unique SWF structure on Y given one on X making it final. For $U \subseteq Y$ open, define

$$\mathcal{O}_Y(U) = \{f: U \rightarrow k \mid f \circ \pi \in \mathcal{O}_X(\pi^{-1}U)\}.$$

Then the coequalizer of two SWF morphisms is given as the SWF on the coequalizer of the topological spaces, which is a quotient.

- (4) From lecture we know that for any SWF $(X, \mathcal{O}_X)$ we have

$$\text{hom}_{\text{SWF}}(X, \mathbb{A}^n) \cong \mathcal{O}_X(X)^n$$

naturally, and thus

$$\text{hom}_{\text{SWF}}(X, \mathbb{A}^n) \cong \text{hom}(X, \mathbb{A}^1)^n$$

naturally. So by Yoneda, $\mathbb{A}^n$ is the n -fold product of $\mathbb{A}^1$ with itself in SWF.

But $\mathbb{A}^2$ is not homeomorphic to $\mathbb{A}^1 \times \mathbb{A}^1$ as topological spaces. The closed sets of $\mathbb{A}^1$ are the finite subsets of $\mathbb{A}^1$ itself, so the closed subsets of $\mathbb{A}^1 \times \mathbb{A}^1$ are all of the form $S_1 \times S_2, S_1 \times \mathbb{A}^1, \mathbb{A}^1 \times S_2, \mathbb{A}^1 \times \mathbb{A}^1$ for S_i finite. So $Z(x^2 + y^2) \subseteq \mathbb{A}^2$ for example cannot be homeomorphic to a closed subset of $\mathbb{A}^1 \times \mathbb{A}^1$.

So the forgetful functor doesn't preserve limits and is thus not a right adjoint.

- (5)

- (6) For the first, let $X = \mathbb{R}$ with two SWFs $(X, \mathcal{O}_1)$ and $(X, \mathcal{O}_2)$. The first being the SWF of smooth maps and the second being the SWF of continuous maps on $\mathbb{R}$. $\text{id}: X \rightarrow X$ is not a morphism of SWFs $\mathcal{O}_1 \rightarrow \mathcal{O}_2$ since this would imply $\mathcal{O}_1 \subseteq \mathcal{O}_2$, which is not true (not all continuous maps are smooth).

Conversely $\text{id}: \mathcal{O}_2 \rightarrow \mathcal{O}_1$ from continuous maps to smooth maps is a morphism of SWFs. It is not an isomorphism since its inverse would have to be $\text{id}: \mathcal{O}_2 \rightarrow \mathcal{O}_1$ (id's inverse is itself) which we already showed is not a morphism of SWFs, while $\text{id}: X \rightarrow X$ is clearly a homeomorphism.

7.3 Problem

Define $U_x = U_y = U_z = \mathbb{A}^2$, and we denote their coordinates as follows:

$$k[U_x] = k\left[\frac{y}{x}, \frac{z}{x}\right], \quad k[U_y] = k\left[\frac{x}{y}, \frac{z}{y}\right], \quad k[U_z] = k\left[\frac{x}{z}, \frac{y}{z}\right].$$

Further define

$$U_{xy} = D_{U_x}(y/x), \quad V_{xy} = D_{U_y}(x/y),$$

and

$$\varphi_{xy}: U_{xy} \rightarrow V_{xy}, \quad (a, b) \mapsto \left(\frac{1}{a}, \frac{b}{a}\right).$$

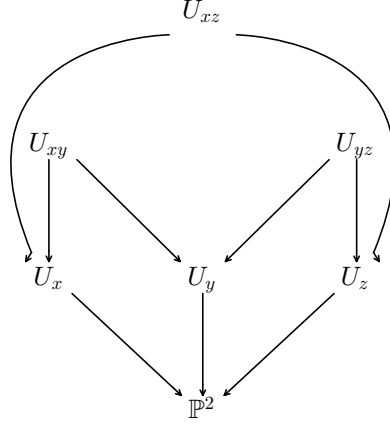
We do the same for the other two pairs of variables.

- (1) Prove that φ_{xy} is a well-defined isomorphism of varieties.
- (2) Via the maps defined, we glue U_x, U_y, U_z together and obtain the **projective plane** $\mathbb{P}^2$. Prove that $\mathcal{O}_{\mathbb{P}^2}(\mathbb{P}^2) = k$ and deduce that $\mathbb{P}^2$ is not affine.
- (3) Find a natural bijection between $\mathbb{P}^2$ and the set of lines through the origin in k^3 . Given a nonzero point $(a, b, c) \in k^3$ we denote the line through it and the origin by $[a : b : c]$. These are called the **projective coordinates** of $\mathbb{P}^2$.
- (4) Let $a, b, c \in k$ such that a or b is nonzero. Consider

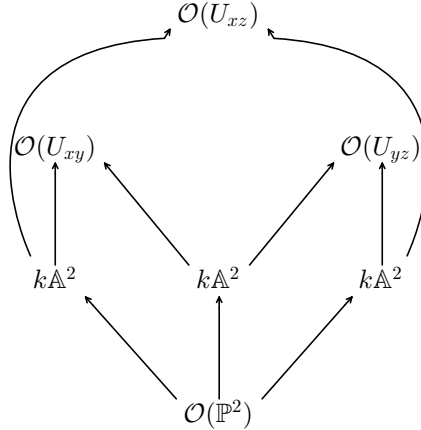
$$Z_{U_z}\left(a\frac{x}{z} + b\frac{y}{z} + c\right) \subseteq U_z.$$

Let L be its image in $\mathbb{P}^2$. Show that $\overline{L}$ has a single point not in L and write its projective coordinates.

- (1) Firstly φ_{xy} lands in V_{xy} since if $0 \neq a = y/x$ then $0 \neq 1/a = x/y$ so $(1/a, b/a)$ is not a zero of x/y . Since U_{xy}, V_{xy} are open subspaces of affine space, it is sufficient to check that φ_{xy} 's components are rational functions to see it is a morphism of SWFs, which they clearly are. And φ_{xy} 's inverse is given by ψ_{xy} defined $V_{xy} \rightarrow U_{xy}$ by $(c, d) \mapsto (1/c, d/c)$. This is symmetric to φ_{xy} and so is also a morphism of SWFs.
- (2) We have the following colimiting diagram,



And since global sections preserves contravariantly preserves colimits, the following diagram is limiting



The map $k\mathbb{A}^2 = \mathcal{O}(U_y) \rightarrow \mathcal{O}(U_{xy})$ is the map $p \mapsto p \circ \varphi_{xy}$, which is the function $(a, b) \mapsto p(1/a, b/a)$. So $\mathcal{O}(\mathbb{P}^2)$ are triplets of polynomials p_1, p_2, p_3 such that

$$p_1 = p_2 \circ \varphi_{xy}, \quad p_1 = p_3 \circ \varphi_{xz}, \quad p_2 = p_3 \circ \varphi_{yz}.$$

But $p_2 \circ \varphi_{xy}$ is not polynomial in (a, b) , unless $p_2 \in k$. So we see that $p_1 = p_2 = p_3 \in k$, meaning $\mathcal{O}(\mathbb{P}^2) = k$.

For an affine variety Y we have $\mathcal{O}(Y) = k[Y] \cong k[\mathbb{A}^n]/\mathcal{I}(Y)$. This is isomorphic to k iff $\mathcal{I}(Y)$ is maximal, i.e. iff $Y = Z \circ \mathcal{I}(Y)$ is a single point. But $\mathbb{P}^2$ is not a single point, as it contains copies of $\mathbb{A}^2$ so it is not affine.

- (3) As a topological space, $\mathbb{P}^2$ is the disjoint union $U_x \amalg U_y \amalg U_z$ quotiented by $(a, b) \in U_x$ being identified with $(1/a, b/a) \in U_y$ and $(1/a, b/a) \in U_z$ for $a \neq 0$ and similar for U_y .

For a line $[a : b : c]$ we map it to the following points:

$$[a : b : c] \mapsto \begin{cases} (b/a, c/a) \in U_x \\ (a/b, c/b) \in U_y, \\ (a/c, b/c) \in U_z \end{cases},$$

when these are defined. Since $(a, b, c) \neq 0$ at least one will be defined. And when two are defined, e.g. $a, b \neq 0$ then we see that

$$(b/a, c/a) \sim (a/b, c/b),$$

so that $[a : b : c]$ determines the same point in $\mathbb{P}^2$.

And for any point in $\mathbb{P}^2$, say $(b, c) \in U_x \subseteq \mathbb{P}^2$ then $[1 : b : c]$ is mapped to it. So this mapping is surjective and injective.

- (4) The line $a\frac{x}{z} + b\frac{y}{z} + c = 0$ can be homogenized to give $ax + by + cz = 0$. So the closure of L is the projective zero set of this homogeneous polynomial:

$$\bar{L} = \{[x : y : z] \mid ax + by + cz = 0\}.$$

When $z \neq 0$ then this is in L , as it is in U_z . And there is a unique solution when $z = 0$: $ax + by = 0$ so the solution is

$$[-b : a : 0].$$

7.4 Problem

Suppose that for every k -algebra A we have a function $f_A: A \rightarrow A$. Further suppose that these commute for all k -algebra morphisms: for all $\varphi: A \rightarrow B$ we have $\varphi \circ f_A = f_B \circ \varphi$. Show that there is a $f \in k[x]$ such that $f_A(a) = f(a)$ for all $a \in A$.

Consider $f_{k[x]}: k[x] \rightarrow k[x]$ and let $f \in k[x]$ be x 's image under $f_{k[x]}$. Then for $a \in A$ let $\varphi: k[x] \rightarrow A$ be the unique k -algebra morphism given by $\varphi(x) = a$. Then we have

$$f_A(a) = f_A \circ \varphi(x) = \varphi \circ f_{k[x]}(x) = \varphi(f(x)),$$

and as a k -algebra morphism φ commutes with polynomials, so $\varphi(f(x)) = f(\varphi(x)) = f(a)$. Thus we have $f_A(a) = f(a)$ as desired.

7.5 Problem

Let $X \subset \mathbb{A}^n$ be a proper affine subvariety over $\mathbb{C}$.

- (1) Let μ be the Lebesgue measure on $\mathbb{A}^n$, show that $\mu(X) = 0$.
- (2) Deduce that “almost all complex matrices are diagonalizable”.
- (3) Assume X is not finite. Show that X is unbounded in the Euclidean metric on $\mathbb{A}^n$ and thus not compact in the Euclidean topology.

- (1) Since X is closed, $X \subseteq Z(f)$ for a polynomial $f \in k[\mathbb{A}^n]$. Let χ denote the indicator function of $Z(f)$, so by Fubini-Tonelli,

$$\mu(X) \leq \mu(Z(f)) = \int_{\mathbb{A}^n} \chi = \int_{\mathbb{A}^1} \cdots \int_{\mathbb{A}^1} \chi(x_1, \dots, x_n) dx_n \cdots dx_1.$$

For $a_1 \in \mathbb{A}^1$ we have that $\chi(a_1, x_2, \dots, x_n)$ is the indicator function of $Z(f(a_1, x_2, \dots, x_n))$.

Now $f(a, \bar{x})$ is identically zero iff $f(y, \bar{x}) = (y-a)g(y, \bar{x})$ for some polynomial g . $g(y, \bar{x})$'s maximal degree in y is less than f 's so there can be only finitely many values a such that $f(a, \bar{x})$ is identically zero. That is, for almost every a we have that $Z(f(a, \bar{x}))$ is a proepr subvariety of $\mathbb{A}^{n-1}$.

So we have

$$\mu(Z(f)) = \int_{\mathbb{A}^1} \mu(Z(f(a, \bar{x}))) da.$$

Since for almost all a , $Z(f(a, \bar{x}))$ is a proper subvariety, by induction on n its measure is zero. So the above integrand is zero almost everywhere, and thus $\mu(Z(f)) = 0$ so $\mu(X) = 0$.

For the base step, we just note that for a single-variable polynomial $f \in k[x]$, $Z(f)$ is finite and thus has zero measure.

- (2) If a matrix A is not diagonalizable, it must in particular have a repeated eigenvalue. That is, its characteristic polynomial p_A has a multiple root. This occurs iff its discriminant is zero, $\text{disc}(p_A) = 0$. Since p_A 's coefficients are polynomial in A and the discriminant is polynomial in its input, $A \mapsto \text{disc}(p_A)$ is polynomial.

So the set of non-diagonalizable matrices is contained in $Z(\text{disc}(p_\bullet))$ which has measure zero, as it is proper (some matrices don't have repeated eigenvalues).

- (3) Let $X = Z(I)$, and consider $A = k[\mathbb{A}^n]/I = k[X]$. Then A is a fg k -algebra and thus by Noether's normalization lemma there is an embedding $k[\mathbb{A}^m] \rightarrow A$. This gives rise to a morphism of SWFs $\mathbb{A}^m \rightarrow X$ which must be surjective, as it is induced by an injective map. Furthermore, this is a continuous map as it is locally rational, which is continuous in the Euclidean topology. Since $\mathbb{A}^m$ is unbound in $\mathbb{A}^n$, so too must X be.